

Infrastructure Implementation Standard: Dual-Tesla P40 Inference Environment

1. System Foundation and Baseline Hardening

In edge and homelab environments, establishing a "security-first" host configuration is not merely a best practice; it is a strategic requirement for protecting proprietary weights and ensuring high-availability uptime. The host operating system serves as the foundational layer of the inference engine's "nervous system." Any compromise here directly threatens the integrity of the data lifecycle and the reliability of the local AI agent.

Hardening the Operating Environment

The mandatory operating system is **Ubuntu Server 22.04 LTS**. A tiered security posture must be implemented to eliminate common attack vectors:

- **Authentication Standard:** SSH access must be restricted to key-only authentication. Password-based logins and root-level SSH access must be explicitly disabled in `sshd_config`.
- **Update Management:** The unattended-upgrades service must be configured to prioritize security patches. To prevent unexpected service interruptions or driver/kernel mismatches during active inference, automatic reboots must be disabled (Automatic-Reboot "false").

Mandatory Firewall (UFW) Policy

The network security posture follows a "Default Deny" incoming stance. Access is restricted to administrative and encrypted overlay interfaces.

Direction	Policy	Interface/Port	Purpose
Incoming	Deny	All	Default security posture
Incoming	Allow	OpenSSH	Administrative access
Incoming	Allow	Tailscale	Encrypted overlay network access
Incoming	Allow	8765/TCP	WonderBot Remote AV Bridge
Outgoing	Allow	All	System updates and model downloads

Critical Warning: Docker's default networking configuration modifies iptables and often bypasses UFW rules. Consequently, all inference APIs and observability services must be bound to 127.0.0.1 or the specific Tailscale IP, never 0.0.0.0. To further mitigate risk, **Fail2ban** must be deployed to monitor and ban brute-force SSH attempts.

2. GPU Subsystem and Thermal Management Standards

The Tesla P40, based on the Pascal architecture, is now classified as "feature-complete and frozen." For infrastructure planning, architects must note the **EOL timeline**: enterprise 580 branch updates will run until the end of July 2026, after which the hardware transitions to security-only maintenance.

Driver Specification and Stability

To prevent kernel-update regressions, the system must utilize the `nvidia-driver-570-server` branch with **DKMS (Dynamic Kernel Module Support)**. Architects must explicitly forbid bleeding-edge non-server drivers. To maintain environmental integrity, administrators must use `sudo apt-mark hold` on the driver and DKMS packages to prevent apt upgrade from breaking the inference stack.

Performance and Thermal Directives

- **Persistence Mode:** This must be enabled to prevent GPUs from dropping power states between inference calls, which leads to unacceptable latency and "vanishing" cards from the PCIe bus.
- **Power Capping:** In shared chassis environments, the P40 (rated at 250W) must be capped at **200W**. This enforces thermal longevity with negligible performance loss.
- **ECC VRAM:** Error Correction Code (ECC) must remain **enabled**. For a high-availability inference box, the minor throughput cost is secondary to the necessity of stability and memory reliability.

Automated Tuning Service

Automated initialization is superior to manual entry. A systemd one-shot service (nvidia-tune.service) must be implemented to ensure power limits and persistence are applied on boot:

```
[Unit]
After=multi-user.target
[Service]
Type=oneshot
RemainAfterExit=true
ExecStart=/usr/bin/nvidia-smi -pm 1
ExecStart=/usr/bin/nvidia-smi -pl 200
[Install]
WantedBy=multi-user.target
```

3. Storage Architecture and Data Lifecycle Policy

Large-scale model tensors (GGUF/HF) require high-throughput storage to avoid OS-level contention. Read-heavy workloads on these files necessitate a dedicated architecture.

Weights SSD Standard

The primary storage is a **1.92TB dedicated SSD**.

- **File System:** Format with ext4.
- **Mounting:** The /etc/fstab entry must include the noatime flag to eliminate pointless write churn of access-time metadata, thereby extending the life of the SSD.

OSError 28 (Space Exhaustion) Mitigation

A critical failure mode identified in dual-Xeon P40 systems is the disparity between the relatively small root partition (~97GB) and the large weights drive. To prevent system lockups during model downloads or tensor shuffling, the following environment variables **must** be redirected to /mnt/weights/tmp:

- TMPDIR, TEMP, and TMP
- HF_HOME, HF_HUB_CACHE, and HF_ASSETS_CACHEThe wofl service user must maintain recursive ownership of the /mnt/weights directory, permitting model management without root intervention.

4. Network Isolation and Remote Access via Tailscale

Exposing unauthenticated inference APIs to the public internet is a catastrophic security risk. Instead, the environment standardizes on **Tailscale** for remote access.

Localhost Binding and Remote AV Bridge

All services—including Ollama, Grafana, and the WonderBot Bridge—must be bound to 127.0.0.1. The "Remote AV Bridge" architecture unbundles the "Brain" (the server) from the "Senses" (the remote client). Remote Windows clients capture camera and microphone data, streaming it to the server via the bridge on port **8765**. This maintains a strict network boundary for the core compute server while allowing it to perceive remote environments.

5. Observability and Health Monitoring Stack

Real-time telemetry is mandatory to identify VRAM bottlenecks and thermal throttling. The observability stack is deployed via Docker Compose, consisting of:

1. **Prometheus:** Time-series database.
2. **Grafana:** Visualization layer.
3. **Node Exporter:** Host metrics.
4. **NVIDIA GPU Exporter:** NVML-based GPU telemetry.

Grafana Dashboard Standard

Dashboards must be configured with specific P40 thresholds:

- **VRAM:** Monitor against the 24GiB hardware limit per card.
- **Power:** Monitor live wattage against the enforced 200W limit.
- **Thermals:** Visual alerts must be set to **Amber at 83°C** and **Red at 88°C**. Administrative credentials must be managed via a .env file to ensure secrets are never committed to version control.

6. Inference Optimization and Runtime Execution

Maximizing throughput on dual-socket Xeon systems requires mitigating cross-socket (QPI/UPI) latency.

NUMA-Awareness Standard

To prevent the performance penalty of threads running on a CPU socket disconnected from the target GPU, architects must:

1. Use `nvidia-smi topo -m` to identify which CPU socket is physically local to each P40.
2. Use `numactl --hardware` to verify the memory nodes.
3. Deploy `numactl` to pin inference processes to the local socket and memory node.

Kernel and Model Selection

- **Low-Swappiness:** Set `vm.swappiness=10` to prevent the kernel from paging model data to disk, which causes severe latency spikes during KV-cache operations.
- **Model Standard:** Prioritize **GGUF/Quantized (Q4/Q5)** models. The P40's architecture lacks modern FP16/Tensor core acceleration; quantized formats leverage its strengths in integer and FP32 math.

Final Summary

By adhering to these standards—enforcing driver pinning, redirecting temporary directories to avoid root exhaustion, and utilizing NUMA-aware process pinning—the dual-P40 environment is transformed into a resilient, architecturally sound inference node. These protocols ensure the system remains secure, observable, and performant throughout the remaining lifecycle of the Pascal architecture.